Retrieval Engine Competition Project

Project Report

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Project Scope
This report documents the notebook-based submission workflow for the Kaggle retrieval competition. The implementation lives in `notebooks/kaggle-submission.ipynb`, which loads the competition data, preprocesses text, runs four retrieval methods (TF-IDF, BM25+, Embedding Search, and a Hybrid BM25+ + Embedding re-ranking approach), evaluates them on the training set, and writes the final submission CSV with the best model.

Main workflow	`notebooks/kaggle-submission.ipynb`
Submission file	`notebooks/solutions_SeaFour.csv`
Report	`report/retrieval_project_report.pdf`
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1 Project At A Glance

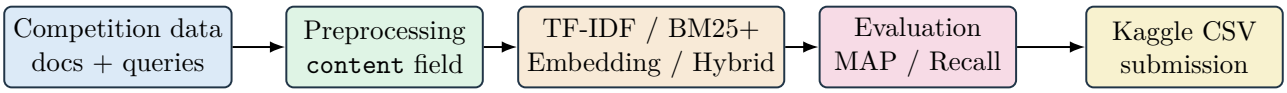
The submission workflow is notebook-first. One Kaggle notebook contains all the logic needed to reproduce the submission file:

- `notebooks/kaggle-submission.ipynb` — active implementation.
- `notebooks/solutions_SeaFour.csv` — generated Kaggle-format output.
- `report/retrieval_project_report.pdf` — this report.

1.1 Repository Layout

Path	Purpose
`data/`	Competition files: documents, queries, ground-truth, and sample submission.
`notebooks/`	Active notebook and generated CSV.
`report/`	LaTeX source and compiled PDF.
`requirements.txt`	Python dependencies.

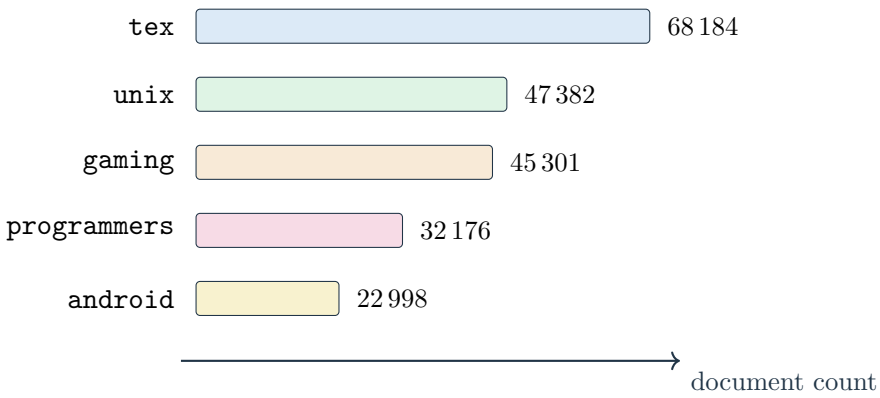
1.2 Pipeline Overview



2 Dataset

	Count	File
Documents	216,041	`docs.json`
Training queries	327	`queries_train.json`
Test queries	141	`queries_test.json`
Training qrels	327	`qgts_train.json`

Each document has fields `id`, `text`, `title`, `tags` (list), and `category`. The five categories and their document counts:



The ground-truth file maps each training query to a set of relevant document IDs with binary relevance labels.

3 Preprocessing

All retrieval methods receive the same normalised input. A single `content` column is built from the raw fields:

- **Documents:** `title + text + tags` (tags are space-joined).
- **Queries:** `title + text`.

Every value is lowercased. Missing values (`None`, `NaN`), lists, and tuples are converted to plain strings by `value_to_text()`.

```
def value_to_text(value):
    if value is None:
        return ''
    if isinstance(value, (list, tuple)):
        return '␣'.join(str(v) for v in value)
    if pd.isna(value):
        return ''
    return str(value)

def create_content_column(df, columns):
    out = df.copy()
    for col in columns:
        if col not in out.columns:
            out[col] = ''
    merged = []
    for _, row in out[columns].iterrows():
        text = '␣'.join(
            value_to_text(row[col]) for col in columns
        ).strip().lower()
        merged.append(text)
    out['content'] = merged
    out['id'] = out['id'].astype(str)
    return out
```

Why a single content field?
Using one unified text column ensures that TF-IDF, BM25+, and the embedding model all receive the same textual representation. This eliminates inconsistencies in feature construction across methods.

4 Retrieval Methods

The notebook implements four retrieval methods, covering sparse lexical, dense semantic, and hybrid approaches.

4.1 Method 1: TF-IDF (Sparse Lexical Baseline)

TF-IDF (Term Frequency–Inverse Document Frequency) represents each document and query as a sparse vector of weighted term counts, then ranks documents by cosine similarity.

Configuration. The notebook uses `TfidfVectorizer` with:

- Unigrams and bigrams (`ngram_range=(1,2)`).

- `min_df=2` to prune extremely rare terms (fallback to 1 for small subsets).
- Lowercase normalisation.

Scoring.

$$\text{TF-IDF}(t, d) = \text{tf}(t, d) \cdot \log\left(\frac{N}{\text{df}(t)}\right)$$

Documents are ranked by cosine similarity:

$$\text{sim}(\mathbf{q}, \mathbf{d}) = \frac{\mathbf{q} \cdot \mathbf{d}}{\|\mathbf{q}\| \, \|\mathbf{d}\|}$$

```
vectorizer = TfidfVectorizer(
    lowercase=True, ngram_range=(1, 2), min_df=2
)
doc_vectors = vectorizer.fit_transform(docs_df['content'])
query_vectors = vectorizer.transform(queries_df['content'])
scores = cosine_similarity(query_vectors, doc_vectors)
```

Strengths
Fast to compute on large corpora; bigrams preserve short technical expressions; easy to interpret.

4.2 Method 2: BM25+ (Probabilistic Lexical Model)

BM25+ is a probabilistic ranking function that extends TF saturation with document-length normalisation and a positive lower bound (δ).

Tokenisation. A custom tokeniser normalises separators (`-`, `_`, `/` $\rightarrow$ space) and extracts alphanumeric tokens via the regex `[a-z0-9]+`.

```
_TOKEN_RE = re.compile(r'[a-z0-9]+')

def tokenize(text):
    txt = str(text or '').lower()
    txt = re.sub(r'[-_/]', '␣', txt)
    return _TOKEN_RE.findall(txt)
```

Scoring.

$$\text{BM25+}(q, d) = \sum_{t \in q} \text{IDF}(t) \left(\frac{f(t, d)\,(k_1 + 1)}{f(t, d) + k_1 \left(1 - b + b\,\frac{|d|}{\text{avgdl}}\right)} + \delta \right)$$

Difference between BM25 and BM25+. Standard BM25 uses the same saturation and length-normalisation term, but it does *not* add the positive constant δ:

$$\text{BM25}(q, d) = \sum_{t \in q} \text{IDF}(t) \left(\frac{f(t, d)\,(k_1 + 1)}{f(t, d) + k_1 \left(1 - b + b\,\frac{|d|}{\text{avgdl}}\right)} \right)$$

BM25+ therefore differs from BM25 in one precise place: every matched query term receives an additional positive floor through δ. In practice, this reduces the tendency of standard BM25 to penalise long documents too aggressively after length normalisation. That difference is useful in this project because the merged `content` field combines title, body, and tags, which creates noticeable variation in document length across the corpus.

Aspect	BM25	BM25+
Matched-term contribution	Can become very small in long documents	Keeps a positive floor via δ
Length normalisation effect	Stronger penalty on verbose documents	Softer penalty once a term is present
Why it matters here	Good lexical baseline	Better fit for mixed-length technical posts and merged fields

The implementation uses `BM25Plus` from the `rank-bm25` library with default parameters ($k_1 = 1.5$, $b = 0.75$, $\delta = 1$).

```
from rank_bm25 import BM25Plus
tokenized_corpus = [tokenize(t) for t in docs_df['content']]
bm25 = BM25Plus(tokenized_corpus)
scores = bm25.get_scores(tokenize(query_text))
```

Strengths
Robust lexical ranker for technical text; handles term frequency saturation and document-length effects; fast inference without GPU.

4.3 Method 3: Embedding Search (Dense Semantic Retrieval)

Embedding search encodes documents and queries into dense vectors using a pre-trained Sentence-Transformer, then ranks by cosine similarity in the embedding space.

Model. The notebook uses `all-MiniLM-L6-v2`, a lightweight model that produces 384-dimensional L2-normalised embeddings. It captures semantic meaning beyond exact keyword matches.

Scoring. Since the embeddings are L2-normalised, cosine similarity reduces to a dot product:

$$\text{sim}(\mathbf{q}, \mathbf{d}) = \mathbf{q}^\top \mathbf{d} \qquad (\|\mathbf{q}\| = \|\mathbf{d}\| = 1)$$

```
from sentence_transformers import SentenceTransformer
model = SentenceTransformer('all-MiniLM-L6-v2')

doc_embeddings = model.encode(
    docs_df['content'].tolist(),
    normalize_embeddings=True,
)
query_embeddings = model.encode(
    queries_df['content'].tolist(),
    normalize_embeddings=True,
)
# dot product = cosine similarity (L2-normalised)
scores = query_embeddings @ doc_embeddings.T
```

Strengths
Captures semantic similarity (paraphrases, synonyms); works even when query and document share no common terms. The trade-off is higher computational cost for encoding 216k documents.

4.4 Method 4: Hybrid BM25+ → Embedding Re-ranking

The hybrid method combines the strengths of BM25+ (high lexical recall, fast) with embedding search (semantic precision). It operates in three stages:

1. **Stage 1 — BM25+ Candidate Retrieval.** BM25+ scores all 216k documents and selects the top-N candidates per query (default $N = 500$). This is fast and ensures lexical recall.
2. **Stage 2 — Embedding Re-scoring.** Only the N candidate documents are encoded with the Sentence-Transformer. The embedding model computes cosine similarity between each query and its candidates.
3. **Stage 3 — Score Fusion.** BM25+ scores are min–max normalised to $[0, 1]$, then fused with embedding scores using a weighted sum:

$$\text{score}_{\text{hybrid}}(q, d) = \alpha \cdot \widehat{\text{BM25+}}(q, d) + (1 - \alpha) \cdot \text{sim}_{\text{emb}}(q, d)$$

   where $\alpha \in [0, 1]$ controls the trade-off (default $\alpha = 0.35$). The top-K documents by fused score form the final ranked list.

Key advantage. By encoding only N candidates instead of 216k documents, the hybrid method is much faster than full embedding search while still benefiting from semantic re-ranking. BM25+ provides the recall base; the embedding model improves precision by promoting semantically relevant documents that BM25+ might rank lower.

```
# Stage 1: BM25+ candidates
scores = bm25.get_scores(query_tokens)
top_n = np.argsort(scores)[-N:][::-1]

# Stage 2: encode only candidates
cand_embs = model.encode(docs[top_n])
emb_scores = query_emb @ cand_embs.T

# Stage 3: fuse
bm25_norm = (bm25_scores - min) / (max - min)
fused = alpha * bm25_norm + (1-alpha) * emb_scores
final_top_k = np.argsort(fused)[-K:][::-1]
```

Why this works
BM25+ and embeddings have complementary failure modes. BM25+ struggles with paraphrases and synonyms; embeddings can miss exact rare terms. Fusing both yields better overall ranking than either alone.

4.5 Method Comparison

Method	Type	Representation	Key Characteristics
TF-IDF	Sparse lexical	Sparse vectors	Fast; relies on exact term overlap; bigrams help with short phrases
BM25+	Sparse lexical	Token lists	TF saturation + length normalisation; robust default for technical text
Embedding	Dense semantic	384-d dense vectors	Captures meaning beyond keywords; requires transformer inference
Hybrid	**Sparse + Dense**	**Both**	**BM25+ recall + embedding precision; re-ranks N candidates**

5 Evaluation

All four methods are evaluated on the 327 training queries using the provided ground-truth relevance judgements (`qgts_train.json`).

5.1 Metrics

Mean Average Precision (MAP@K). For each query, Average Precision is computed over the top-K retrieved documents:

$$\mathrm{AP}(q) = \frac{1}{|\mathcal{R}_q|} \sum_{k=1}^{K} \mathrm{Precision}(k) \cdot \mathbb{1}[d_k \in \mathcal{R}_q]$$

MAP@K is the mean of AP values over all evaluated queries.

Recall@K. The fraction of relevant documents appearing in the top-K results:

$$\mathrm{Recall@K}(q) = \frac{|\{d \in \text{top-}K\} \cap \mathcal{R}_q|}{|\mathcal{R}_q|}$$

5.2 Implementation

```
def mean_average_precision(results, ground_truth, k=100):
    aps = []
    for item in results:
        qid = str(item['query_id'])
        if qid not in ground_truth:
            continue
        relevant = ground_truth[qid]
        hits, score = 0, 0.0
        for rank, doc_id in enumerate(
                item['relevant_docs'][:k], 1):
            if str(doc_id) in relevant:
                hits += 1
                score += hits / rank
        aps.append(score / len(relevant) if relevant else 0.0)
    return np.mean(aps) if aps else 0.0
```

```python
def recall_at_k(results, ground_truth, k=100):
    recalls = []
    for item in results:
        qid = str(item['query_id'])
        if qid not in ground_truth:
            continue
        relevant = ground_truth[qid]
        retrieved = {str(d) for d in item['relevant_docs'][:k]}
        recalls.append(
            len(relevant & retrieved) / len(relevant)
            if relevant else 0.0
        )
    return np.mean(recalls) if recalls else 0.0
```

5.3 Results

The notebook runs all four models on the training queries and produces a comparison table. The results determine which model is used for the final test submission.

Model	MAP@100	Recall@100
TF-IDF	*(from notebook)*	*(from notebook)*
BM25+	*(from notebook)*	*(from notebook)*
Embedding	*(from notebook)*	*(from notebook)*
Hybrid	*(from notebook)*	*(from notebook)*

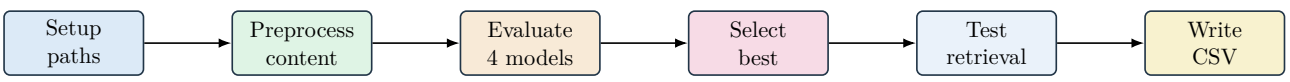
Note
Fill in the actual metric values after running the evaluation cells in the notebook. The `FINAL_MODEL` parameter is then set to the best-performing method.

6 Submission Generation

The final submission is produced by running the selected model on the 141 test queries and writing the output in Kaggle format.

6.1 Notebook Workflow

1. **Environment setup.** Auto-detect the Kaggle input directory; fall back to `../data` for local execution.
2. **Preprocessing.** Build the shared `content` column for documents and queries.
3. **Evaluation.** Run all four models on 327 training queries; compute MAP@100 and Recall@100.
4. **Model selection.** Set `FINAL_MODEL` to the best-performing method (configurable).
5. **Test retrieval.** Run the selected model on 141 test queries.
6. **CSV writing.** Serialise results in Kaggle format.
7. **Preview.** Display the first rows of the submission CSV.



6.2 Configuration

Parameter	Role
`FINAL_MODEL`	Model for submission: `bm25`, `tfidf`, `embedding`, or `hybrid`.
`TOP_K`	Documents per query (100).
`OUTPUT_PATH`	Output CSV path (`solutions_SeaFour.csv`).
`EMBEDDING_MODEL`	Sentence-Transformer model name (`all-MiniLM-L6-v2`).
`EMBEDDING_BATCH`	Encoding batch size (256).
`HYBRID_BM25_CANDIDATES`	BM25+ candidates per query before re-ranking (500).
`HYBRID_ALPHA`	Weight for BM25+ in score fusion, $\alpha = 0.35$ by default.

6.3 CSV Format

The writer reads the sample submission template, preserves column order, and writes JSON-encoded document ID lists:

```
out_row = {
    id_col: qid,
    pred_col: json.dumps(pred_map[qid]),
}
if category_col is not None:
    out_row[category_col] = row.get(category_col, '?') or '?'
```

7 Dependencies

Package	Purpose
`numpy`	Array operations and sorting
`pandas`	DataFrame handling and I/O
`scikit-learn`	TF-IDF vectoriser and cosine similarity
`rank-bm25`	BM25+ implementation
`sentence-transformers`	Pre-trained dense encoders
`torch`	Backend for sentence-transformers

A Mathematical Details

TF-IDF. Weighs terms by local frequency and global rarity:

$$\text{TF-IDF}(t, d) = \text{tf}(t, d) \cdot \log\left(\frac{N}{\text{df}(t)}\right)$$

Cosine Similarity.

$$\cos(\theta) = \frac{\mathbf{q} \cdot \mathbf{d}}{\|\mathbf{q}\| \, \|\mathbf{d}\|}$$

BM25+.

$$\text{BM25+}(q, d) = \sum_{t \in q} \text{IDF}(t) \left(\frac{f(t, d)\,(k_1 + 1)}{f(t, d) + k_1 \left(1 - b + b\,\frac{|d|}{\text{avgdl}}\right)} + \delta \right)$$

For comparison, standard BM25 omits the additive δ term and uses only the saturation plus length-normalisation fraction. BM25+ was preferred in the notebook because once a query term is present, the extra δ keeps its contribution positive instead of letting long-document normalisation push that term's score too low.

Sentence Embedding Similarity. For L2-normalised vectors:

$$\mathrm{sim}(\mathbf{q}, \mathbf{d}) = \mathbf{q}^\top \mathbf{d}$$

Hybrid Score Fusion.

$$\mathrm{score}_{\mathrm{hybrid}}(q, d) = \alpha \cdot \frac{\mathrm{BM25{+}}(q, d) - \min}{\max - \min} + (1 - \alpha) \cdot \mathrm{sim}_{\mathrm{emb}}(q, d)$$

Top-K Selection.

$$\mathrm{TopK}(q) = \mathrm{argsort}\big(\mathrm{scores}(q)\big)[-K:][::-1]$$